

The Sample Analysis Laboratory (SAL) at the German Aerospace Center (DLR) Berlin – a cutting edge laboratory for extraterrestrial material analysis

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ABSTRACT

Sample return missions provide the ultimate ground truth for interpretation of remote sensing data and enable an unprecedented level of analysis compared to in-situ instruments. In response to the explosion of interest in this field in recent years the German Aerospace Center (DLR) in Berlin is constructing a Sample Analysis Laboratory (SAL) for the analysis and curation of returned samples in close cooperation with the Museum für Naturkunde (MfN), and in collaboration with colleagues at NASA and JAXA. Key topics to which SAL will contribute are the geochemical history of hydration, oxidation and alteration of planetary surfaces through analysing natural and artificial samples; detection of organics and possible traces or signs of extinct life; the formation and evolution of planetary bodies. In addition, SAL will extend the capabilities of the existing on-site Planetary Spectroscopy Laboratory facilities with complementary instrumentation for material analysis, supporting preparation for interpretation of NIR and MIR data from instruments on board the VERITAS, EnVision and BepiColombo missions as well as sample curation in the context of the MMX mission.

The laboratory will provide a cleanroom environment of approximately 80 m². The major analytical instruments have already been purchased for SAL: a JEOL iHP200F Field Emission Electron Microprobe Analyzer, a JEOL JSM-IT800 Field Emission Scanning Electron Microscope and a Malvern Panalytical Empyrean X-ray Diffraction system. A vis-IR-microscope (Bruker Hyperion 2000) interfaced with a Bruker Vertex 80V FTIR spectrometer is already in operation and has been used for measurements on Ryugu grains from the Hayabusa2 mission, and spectral analysis of the Ribbeck aubrite. The SAL cleanroom construction is planned to be finished in January 2026, with operation and commissioning due to be well under way by Spring of 2026. This contribution will give an overview of the DLR Sample Analysis Laboratory, the time plan and major goals for the coming years.

Keywords: Extraterrestrial sample return, asteroids, infrared spectroscopy, microscopy, planetary research.

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1. INTRODUCTION

The field of extraterrestrial sample return is growing in importance in the planetary science community. With the substantial advances of the past few decades, robotic sample return missions have brought samples from asteroids Itokawa [1], Ryugu [2] and most recently Bennu [3], from comet Wild 2 [4] and in the near future will bring samples from the Martian moon Phobos [5] and possibly from Mars itself [6]. Returned samples provide the ultimate ground truth to better understand and interpret remote sensing data and allow for far more detailed analysis of extraterrestrial material than what can be achieved using in-situ instruments. In order to preserve pristine returned material special protocols for sample storage, manipulation and transport are required.

In response to the rapid expansion of interest in this area in recent years, a new Sample Analysis Laboratory (SAL) is being constructed at the German Aerospace Center (DLR) in Berlin. This new facility will enable storage and manipulation of sensitive material under a protective dry nitrogen atmosphere. A range of microanalytical instruments in the same laboratory space will allow on-site analysis of material without requiring transfer out of the clean room, minimising possible contamination sources. The new facility will complement the existing Planetary Spectroscopy Laboratory (PSL) and Planetary Analogue Simulation Laboratory (PASLAB), which offer facilities for spectral and micro-spectral characterisation of samples from the UV/visible to near and mid-infrared wavelength range, as well as analysis with Raman spectroscopy and digital 3D microscopy [7]. In addition, the capabilities of the sample preparation laboratory are being extended to allow preparation of sample standard blocks and thin sections.

The goal of SAL is to provide a comprehensive range of instrumentation for structural, geochemical, mineralogical and elemental analysis at one location in a clean room environment for the analysis of extraterrestrial material in the form of returned samples and meteorites, allowing efficient analysis with a minimal disturbance to sensitive material. The medium term aim of SAL is to establish a European center for extraterrestrial sample curation and analysis in Berlin in close cooperation with the Museum für Naturkunde (MfN). We plan on acting as a sample curation facility for MMX sample allocated to Europe, along with CNES [8].

Key topics that SAL will contribute to are the formation and evolution of planetary bodies, detection of organics and the water history of samples as well as possible traces or signs of extinct life [7].

2. THE LABORATORY LAYOUT AND INSTRUMENTATION

The Sample Analysis Laboratory (SAL) aims to serve the dual purpose of sample curation, as well as sample preparation and analysis within the same laboratory space. The main analytical instruments already purchased for the laboratory are: a field-emission scanning electron microscope (FE-SEM), a field-emission electron microprobe analyser (FE-EMPA), an X-ray diffractometer (XRD), a Fourier Transform Infrared microscope, and a digital microscope. Figure 1 shows the layout of the instruments in the approximately 80 m² ISO 8 clean room space. The technical equipment for the analytical devices (power supply, water cooling, vacuum pumps, etc.) will be housed in the basement directly below the cleanroom to minimise contamination and improve the working environment. Sealed feedthroughs in the floor will be used to connect the instruments to the supplies. Figure 1 shows the layout of the clean room space with the positions of the main instruments and the gloveboxes.

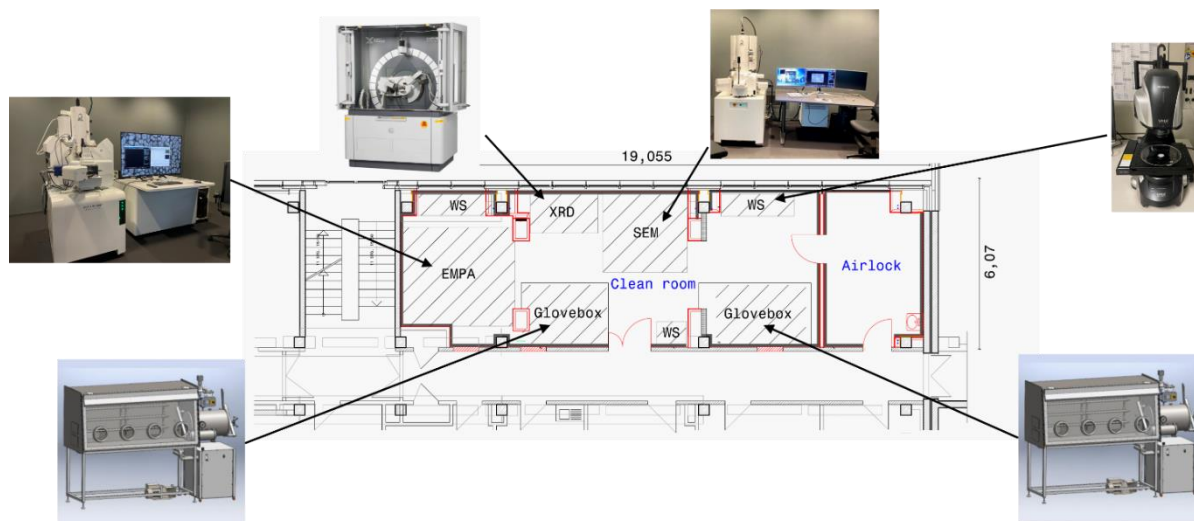


Figure 1 The layout of the Sample Analysis Laboratory with key instruments indicated. Hatched areas represent the service space requirement for the instruments, not the instrument footprints. The double door into the cleanroom is only for initial instrument installation, after which access will be through the airlock only.

The FE-EMPA is equipped with 5 wavelength dispersive x-ray spectrometers (WDS) with the interchangeable crystals TAPL/J, LDE1L/2, PETL/J/H and LIF/L/H, enabling a comprehensive coverage of elements and flexibility and efficiency in elemental mapping. An energy dispersive x-ray spectrometer (EDS) is also installed for rapid elemental mapping and identification of target areas. The FE-SEM is equipped with an EDS detector for non-destructive rapid compositional analysis, and elemental mapping of samples. An additional backscattered electron (BSE) detector enables high-resolution compositional imaging, while the secondary electron (SE) detector allows high-resolution imaging for detailed surface topographical measurements. The XRD has Bragg-Brentano geometry, which can be switched to parallel beam geometry, and a Cu α source with a 1Der detector. The sample form is flexible, from powders to larger solid objects, also allowing non-destructive measurements of returned samples. Localised μ -XRD analyses can be performed by reducing the beam spot size down to 140 μ m. All these instruments have dedicated sealed sample transport vessels and/or shuttle mechanisms allowing samples to be transferred for measurements from the highly controlled glovebox environment without interaction with the laboratory atmosphere.

For spectral mapping of small samples, a Fourier Transform vis-IR-microscope (Bruker Hyperion 2000) connected with a Bruker Vertex 80V spectrometer will be available, allowing spectral measurements into the mid-infrared (Figure 2). The system has an automated X,Y,Z stage and allows measurements down to a spatial resolution of 50 μ m. This instrument is a key part of the facility, enabling initial spectral characterisation of samples with minimal sample manipulation. Local infrared spectra enable identification of minerals and regions of interest. The system is already in operation and has been used for measurements on returned samples from Ryugu, see Section 4. A Keyence VHX-7000 3D microscope has also been procured and is in operation in existing laboratory facilities, see Figure 2. The microscope allows rapid detailed imaging of sample surfaces, including uneven surfaces, by automated stitching of images in different focal planes, resulting in a quasi-3D representation of samples. This allows quick estimation of sample sizes as well as the dimensions of larger surface features. Sealed transport vessels for the two microscopes are also available.

Two ISO-5 glove boxes are planned, one for sample storage and one for sample preparation. The glove box requirements are currently being defined based on input from partners at NASA JSC and JAXA ISAS. In addition to the reduction of particle contamination, a key requirement is the reduction of volatile contamination sources by using low outgassing vacuum compatible components such as Viton seals and Hypalon gloves for sample manipulation. The glove boxes will be filled with a dry nitrogen environment and active oxygen, humidity and particle measurements will allow the internal environment to be monitored. The gloveboxes will also have a visible microscope installed for initial sample characterizations with minimal disturbance.



Figure 2 The Bruker Hyperion 2000 microscope and the Keyence digital microscope.

3. CURRENT RESEARCH

Whilst the construction of the SAL cleanroom is ongoing, research is already being conducted on samples from the Hayabusa2 mission to the asteroid Ryugu. Work is focused on the effect of space weathering, specifically the geochemical and spectral changes in the infrared within the microcraters and melt splashes. In the context of this work, soon to be published, sample storing and handling procedures have been refined. Samples are stored in a provisional vacuum storage unit (Figure 3a)) and handled as far as possible in a glovebox environment under a dry oxygen-free nitrogen. Figure 3b) and c) shows sample A0383 being transferred from the JAXA facility-to-facility-transfer container (FFTC) into a measurement stage for FTIR microscopic measurements in the glovebox. Figure 3d) and e) show images from the initial inspection of Ryugu sample A0383 performed with the Keyence digital microscope inside the FFTC.

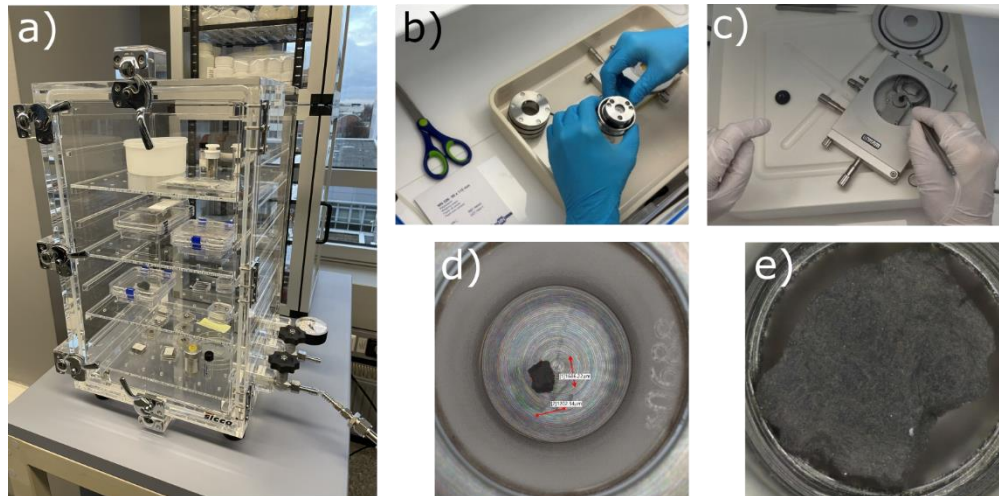


Figure 3 a) Vacuum sample storage unit. b) Opening the JAXA facility-to-facility-transfer container (FFTC) with sample A0383. c) Transferring sample A0383 to the Linkam stage for measurements with the FTIR microscope. d) Keyence microscope images of sample A0383 inside the FFTC e) a higher resolution image of sample A0383 inside the Linkam measurement stage.

A cooperation agreement between the DLR Institute of Space Research and the Museum für Naturkunde (MfN) allows the mutual use of analytical facilities. Until the SAL clean room with its microanalytical instruments is operational, high resolution SEM images of the DLR-Ryugu samples have been made at the MfN facilities. Figure 4 shows example SEM backscattered electrons (BSE) images of regions on sample A0383.



Figure 4. SEM images of sample A0383 taken at the Museum für Naturkunde (MfN) in Berlin.

Micro-FTIR reflectance point measurements were conducted using a Hyperspectral Bruker Hyperion 2000 microscope, coupled to a Bruker Vertex 80V FTIR spectrometer at PSL (DLR). These measurements allow the spectral characterization of samples at the microscale in the VNIR to MIR spectral ranges. Three Ryugu grains (A0112, A0383, and C0404) were analyzed through localized point measurements across their surfaces. This allowed the geochemical analysis of the

samples, identify the effects of space weathering, and enables direct comparison with remote sensing data, such as the NIRS3 spectral dataset. Figure 5 shows an example of a point measurement on the matrix material of Ryugu grain A0383 in the MIR range (2–7 μm).

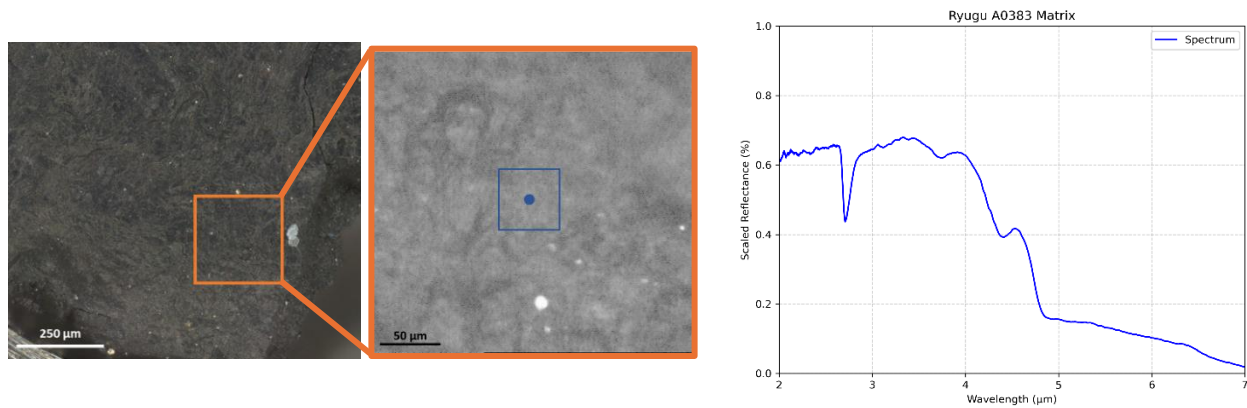


Figure 5. MicroFTIR point measurement of sample A0383 showing the corresponding MIR (2 – 7 microns) reflectance spectrum. The measurement location is marked by a blue dot, and the orange box shows the zoomed-in region of the Ryugu grain A0383.

4. CONSTRUCTION STATUS AND OUTLOOK



Figure 6. The current status of cleanroom construction. Clockwise from bottom left: a view from the airlock into the lab space; lab space with XRD support cutouts; lab space with EMPA support cutouts; a view into the airlock. Cleanroom construction is due to be finished in January 2026.

Figures 6 and 7 show the building progress of the cleanroom space at the point of writing this paper. The SAL is being constructed on the ground floor of the DLR Institute of Space Research in Berlin. The instruments are decoupled from the

floor and are positioned on separate supports. Next steps are installation of the ducts for the air-filtration system, electrical connections, and ESD flooring.



Figure 7. The air filtration system is being installed in the basement. The technical equipment for the analytical devices (vacuum pumps, power supplied, water cooling etc.) will also be installed in the basement.

The Sample Analysis Laboratory cleanroom is due to be finished in January 2026. Following this will be a period of approximately 6 months for installation and commissioning of the main analytical instruments. To develop the facility further as a curation facility will require some extra steps, including procuring the gloveboxes and refining sample storage and handling procedures. We plan to be ready to operate as a curation facility from 2027 onwards and act as a center in Europe for extraterrestrial sample handling training. A major medium-term goal for SAL is to curate a portion of the MMX samples allocated to Europe. The MMX mission is planned to launch in 2026, with samples due to return to Earth in 2031. After an initial phase of analysis and documentation of the samples at JAXA, a portion of the samples is planned to be distributed to Europe. To prepare for this we are in close contact with partners at JAXA and NASA curation centers to gain invaluable experience and train in handling small and sensitive samples, as well as with CNES and ESA to develop a European curation concept for the MMX samples.

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